Introduction to Cloud Computing

Note:

1. The following contents are indicative and just for quick reference.
2. You shall write elaborately in the exam with proper examples/contexts/scenarios.
3. Prepare a business context / scenario for every concept.

MODULE 5

1. DevOps on the Cloud: Continuous Integration/Continuous Deployment (CI/CD) pipelines.

- DevOps is a cultural, technological, and procedural approach that unites software development (Dev) and IT operations (Ops) teams to accelerate the delivery of high-quality, reliable applications. By utilizing automation tools and CI/CD practices, it breaks down departments to enable faster, frequent, and more stable software releases.
- The goals of DevOps are:
  - Faster and continuous software releases.
  - Reduces manual errors through automation.
  - Built-in monitoring detects failures or issues during delivery or in production.
  - Automate testing throughout the software delivery process.

DevOps – Principles

- **Continuous Integration:**
  - Developers integrate code changes into the source code repository several times per day.
  - Integration is done automatically with tests being conducted on each commit in order to detect conflicts.
  - Makes the code aligned more frequently and helps in maintaining the quality of the code.
- **Continuous Delivery**:
  - Once integrated, the code is always in a deployable state by using the Git commit and push commands.
  - This stage prepares each build for possible release, integrating the development merges with production readiness.
  - Specifically, the pipeline insists that every update can be released as soon as possible if it is of high quality.
- **Continuous Deployment**:
  - In some teams, changes can be directly released into production when the tests have been completed.

- This approach makes the feedback loop narrow and minimizes the deployment overhead.
- While not required for all, continuous deployment is one of the examples of extreme automation in the DevOps approach to achieve zero-latency updates.

DevOps Practice

1. Foster a culture of collaboration and blameless communication
2. Adopt Continuous Integration and Delivery (CI/CD)
3. Set up automated testing
4. Focus on observability and find the right metrics
5. Avoid manual work with automation
6. Incorporate security early in the development lifecycle
7. Learn from incidents and build processes around them
8. Focus first on concepts, then find the right tools
9. Embrace Infrastructure as Code (IaC) and push for a self-service infrastructure
10. Don’t create separate DevOps team
11. Avoid having the DevOps hero
12. Don’t attempt to automate and change everything at once
13. Avoid chasing new tools
14. Don’t sacrifice quality for speed
15. Don’t give up on Continuous Improvement
16. Don’t neglect documentation and information sharing

2. Cloud Migration Strategies

Cloud Migration is the Process of moving data, applications, and other business elements from on-premises infrastructure or traditional data centers to cloud environment like Amazon Web Services(AWS), Microsoft Azure, or Google Cloud. It allows businesses to leverage cloud computing's scalability, flexibility, and cost-effectiveness.

This migration can involve a variety of activities, including moving data storage, servers, applications, and even entire IT infrastructures to the cloud. It is a crucial step for businesses looking to modernize their IT systems and take full advantage of the cloud's benefits, such as real-time processing, improved scalability, and on-demand resources.

Why Cloud Migration is Important

Cloud migration is vital for businesses aiming to improve agility, reduce costs, and enhance their IT infrastructure. By migrating to the cloud, businesses can:

- **Enhance Flexibility**: Cloud platforms provide on-demand resources, allowing businesses to scale quickly based on need.
- **Improve Cost Efficiency**: Cloud services often reduce the need for large upfront investments in hardware and infrastructure, offering **pay-as-you-go** pricing models.
- **Boost Performance and Reliability**: Cloud platforms provide **high availability** and **disaster recovery** solutions to ensure business continuity.
- **Accelerate Innovation**: The cloud offers advanced services like **machine learning**, **big data analytics**, and **AI**, enabling companies to innovate faster.

There are **several cloud migration strategies**, depending on the needs and objectives of the business. Here are seven common strategies:

1. Rehosting (Lift and Shift)

Rehosting, also known as "Lift and Shift," involves taking an existing application and migrating it directly to the cloud, typically using **Infrastructure as a Service (IaaS)**. This method involves minimal changes to the application and is the quickest way to move workloads to the cloud.

2. Replatforming

In this approach, applications are moved to the cloud with slight modifications to optimize them for cloud environments. This may involve upgrading the underlying platform or changing how the application interacts with storage and compute resources, while keeping the overall application architecture the same.

3. Repurchasing

Repurchasing involves replacing legacy applications with cloud-based Software-as-a-Service (SaaS) solutions. This strategy is often used when the current on-premises application is outdated, and a cloud-native solution offers better functionality, security, and scalability.

4. Refactoring

Refactoring refers to reusing the application code and frameworks while adapting it to run more efficiently on a Platform as a Service (PaaS). This strategy optimizes the application for the

cloud, often by leveraging cloud-native features while keeping most of the existing codebase intact.

5. Retiring

Retiring involves decommissioning or discontinuing applications that are no longer needed or useful to the business. This strategy is typically applied to outdated or redundant applications that don't add value and can be removed from the IT environment.

6. Retaining

In some cases, businesses choose to keep certain applications on-premises, while migrating others to the cloud. Retaining applications on-premises might be necessary due to regulatory requirements, cost considerations, or because they are not yet compatible with cloud environments.

7. Hybrid Cloud Migration

Hybrid migration involves using both on-premises infrastructure and cloud services to create a mixed environment. This approach offers flexibility by allowing businesses to maintain certain workloads on-premises while moving others to the cloud, helping them gradually transition to the cloud or meet specific business needs.

3. Introduction to Big data

What is big data?

- Big data refers to massive, complex data sets that traditional data management systems cannot handle. When properly collected, managed and analyzed, big data can help organizations discover new insights and make better business decisions.
- While enterprise organizations have long collected data, the arrival of the internet and other connected technologies significantly increased the volume and variety of data available, birthing the concept of "big data."
- Today, businesses collect large amounts of data—often measured in terabytes or petabytes—on everything from customer transactions and social media impressions to internal processes and proprietary research.
- Over the past decade, this information has fueled digital transformation across industries. In fact, big data has earned the nickname "the new oil" for its role driving business growth and innovation.
- Data science and, more specifically, big data analytics help organizations make sense of big data's large and diverse data sets. These fields use advanced tools such as machine learning to uncover patterns, extract insights and predict outcomes.

- In recent years, the rise of artificial intelligence (AI) and machine learning has further increased the focus on big data. These systems rely on large, high-quality datasets to train models and improve predictive algorithms.

The difference between traditional data and big data

- Traditional data and big data differ mainly in the types of data involved, the amount of data handled and the tools required to analyze them.
- Traditional data primarily consists of structured data stored in relational databases. These databases organize data into clearly defined tables, making it easy to query using standard tools like SQL. Traditional data analytics typically involves statistical methods and is well-suited for datasets with predictable formats and relatively small sizes.
- Big data, on the other hand, encompasses massive datasets in various formats, including structured, semi-structured and unstructured data. This complexity demands advanced analytical approaches—such as machine learning, data mining and data visualization—to extract meaningful insights. The sheer volume of big data also requires distributed processing systems to handle the data efficiently at scale.

The V's of big data

The "V's of Big Data" volume, velocity, variety, veracity and value are the five characteristics that make big data unique from other kinds of data. These attributes explain how big data differs from traditional datasets and what's needed to manage it effectively.

Volume: Big data is "big" because there's more of it. The massive amount of data generated today—from web apps, Internet of Things (IoT) devices, transaction records and more—can be hard for any organization to manage. Traditional data storage and processing systems often struggle to handle it at scale. Big data solutions, including cloud-based storage, can help organizations store and manage these ever-larger datasets and make sure valuable information is not lost to storage limits.

Velocity: Velocity is the speed at which data flows into a system and big data moves quickly. Today, data arrives faster than ever, from real-time social media updates to high-frequency stock trading records. This rapid data influx provides opportunities for timely insights that support quick decision-making. To handle this, organizations use tools like stream processing frameworks and in-memory systems to capture, analyze and act on data in near real-time.

Variety: Variety refers to the many different formats that big data can take. Along with traditional structured data, big data can include unstructured data, such as free-form text,

images and videos. It can also include semi-structured data, such as JSON and XML files, that have some organizational properties but no strict schema. Managing this variety requires flexible solutions like NoSQL databases and data lakes with schema-on-read frameworks, which can store and integrate multiple data formats for more comprehensive data analysis.

Veracity: Veracity refers to the accuracy and reliability of data. Because big data comes in such great quantities and from various sources, it can contain noise or errors, which can lead to poor decision-making. Big data requires organizations to implement processes for ensuring data quality and accuracy. Organizations often use data cleaning, validation and verification tools to filter out inaccuracies and improve the quality of their analysis.

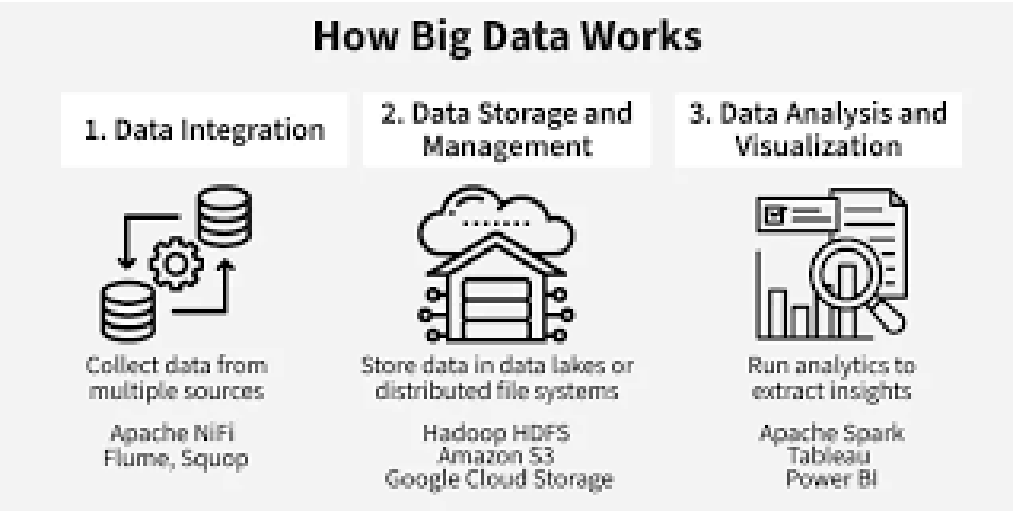
Value: Value refers to the real-world benefits organizations can get from big data. These benefits include everything from optimizing business operations to identifying new marketing opportunities. Big data analytics is critical for this process, often relying on advanced analytics, machine learning and AI to transform raw information into actionable insights.

Below are some of big data's most significant benefits and use cases.

- **Improved decision-making**: Analyzing vast datasets allows organizations to uncover patterns and trends that lead to more informed decisions. For instance, a grocery chain can use sales data and weather forecasts to predict demand for seasonal products, helping to stock stores accordingly and reduce waste.
- **Enhanced customer experience**: Big data enables companies to understand customer behavior at a more granular level, paving the way for more tailored interactions. For instance, big data analytics can help identify customers who frequently buy skincare products from a specific brand. The brand can use this information to help target campaigns for limited-time sales or special offers on similar products.
- **Increased operational efficiency**: Real-time data allows organizations to streamline operations and reduce waste. In manufacturing, for example, organizations can analyze real-time sensor data to predict equipment failures before they occur. This process, known as predictive maintenance, can help prevent downtime and reduce maintenance costs.
- **Responsive product development**: Big data insights help companies respond customer needs and guide product improvements. For example, if multiple users report that a specific feature in a smartphone drains battery life too quickly, developers can prioritize optimizing that feature in the next software update.
- **Optimized pricing**: Big data enables organizations to refine pricing strategies based on real-time market conditions. For example, an airline can use insights derived from

big data to adjust ticket prices dynamically, responding to demand shifts and competitor pricing.

- **Enhanced risk management and fraud detection**: Big data allows organizations to identify and monitor risks proactively. Banks, for instance, analyze transaction patterns to detect potential fraud. If a customer's credit card is used for an unusual high-value purchase in another country, the bank can flag the transaction and notify the customer for verification.
- **Healthcare innovation:** Healthcare providers can use big data to make sense of patient records, genetic information and data from wearable devices. For example, a continuous glucose monitor for a diabetic patient can track blood sugar levels in real-time, allowing healthcare providers to detect dangerous spikes or drops and adjust treatment plans accordingly.



Challenges of big data

While big data offers immense potential, it also comes with significant challenges, especially around its scale and speed. Some of the biggest challenges of big data include:

- **Data quality and management:** Connecting datapoints and keeping data accurate can be a complex undertaking especially with massive amounts of information constantly streaming in from social media, IoT devices and other sources. For example, a logistics company may struggle to integrate GPS data from its fleet with customer feedback and warehouse inventory to get a precise view of delivery performance.
- **Scalability:** As data grows, organizations must expand storage and processing systems to keep up. For instance, a streaming platform analyzing millions of daily viewer interactions may need to constantly add to its storage and compute power to handle demand. Cloud services can offer more scalable alternatives to on-premises solutions, but managing high volumes and velocities of data can still be difficult.

- **Privacy and security**: Regulations like GDPR and HIPAA require strict data privacy and security measures, such as strong access controls and encryption to prevent unauthorized access to patient records. Complying with these mandates can be tough when datasets are massive and constantly evolving.
- **Integration complexity:** Combining different types of data from multiple sources can be technically demanding. For instance, a retail chain may struggle to consolidate structured sales records with unstructured customer reviews and semi-structured supplier data for a comprehensive view of product performance.
- **Skilled workforce:** Big data work requires specialized skills in data science, engineering and analytics. Many organizations face ongoing challenges finding professionals like data analysts and other specialists who can manage and interpret large datasets. For example, a financial institution might struggle to hire data scientists skilled in both machine learning and financial modeling to analyze transaction data and predict market trends.

4. Introduction to Machine Learning

What is machine learning?

Machine learning is the subset of artificial intelligence (AI) focused on algorithms that can "learn" the patterns of training data and, subsequently, make accurate *inferences* about new data. This pattern recognition ability enables machine learning models to make decisions or predictions without explicit, hard-coded instructions.

How machine learning works

Machine learning works through mathematical logic. The relevant characteristics (or "features") of each data point must therefore be expressed numerically, so that the data itself can be fed into a mathematical algorithm that will "learn" to map a given input to the desired output.

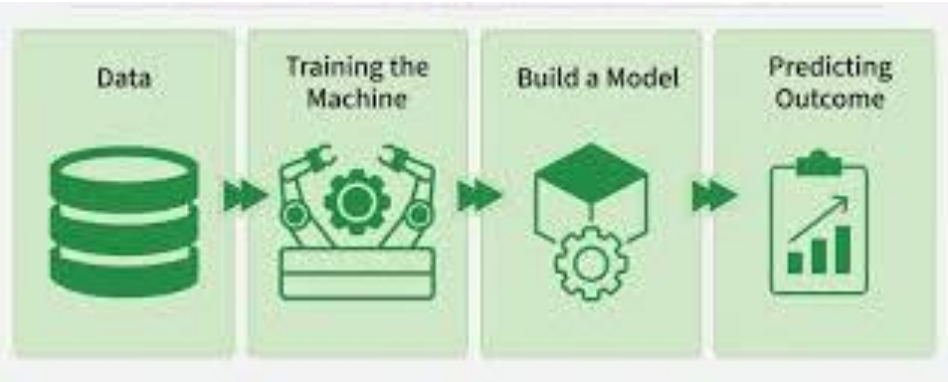
Data points in machine learning are usually represented in vector form, in which each element (or dimension) of a data point's vector embedding corresponds to its numerical value for a specific feature. For data modalities that are inherently numerical, such as financial data or geospatial coordinates, this is relatively straightforward. But many data modalities, such as text, images, social media graph data or app user behaviors, are not inherently numerical, and therefore entail less immediately intuitive feature engineering to be expressed in an ML-ready way.

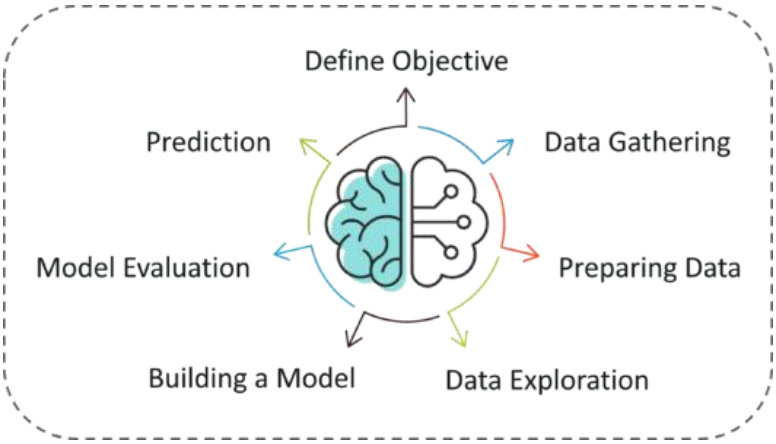
The (often manual) process of choosing which aspects of data to use in machine learning algorithms is called feature selection. Feature extraction techniques refine data down to only

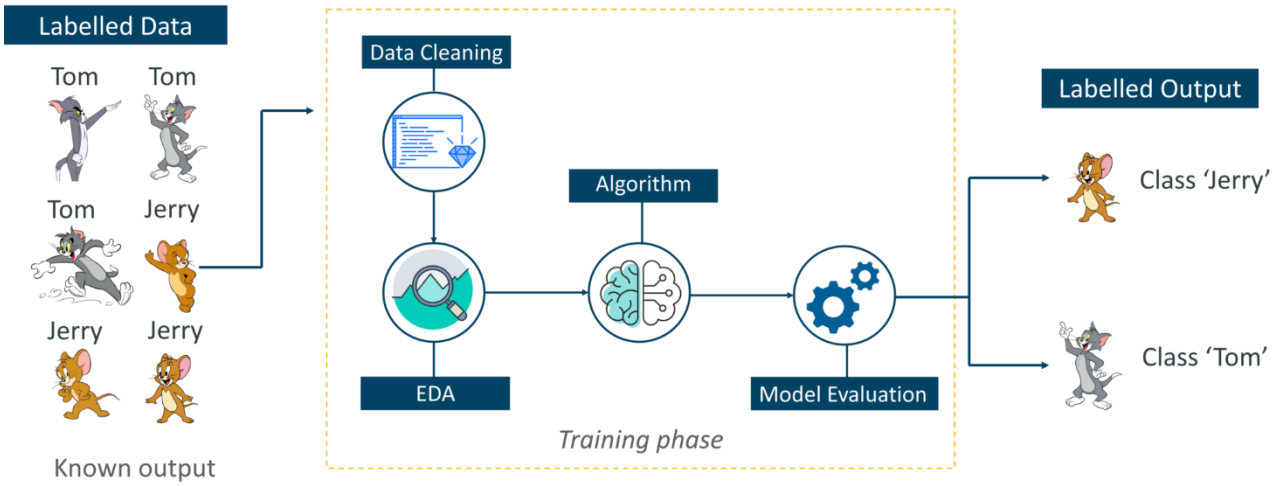
its most relevant, meaningful dimensions. Both are subsets of feature engineering, the broader discipline of preprocessing raw data for use in machine learning. One notable distinction of deep learning is that it typically operates on raw data and automates much of the feature engineering—or at least the feature extraction—process. This makes deep learning more scalable, albeit less interpretable, than traditional machine learning.

Types of machine learning

- All machine learning methods can be categorized as one of three distinct learning paradigms: supervised learning, unsupervised learning or reinforcement learning, based on the nature of their training objectives and (often but not always) by the type of training data they entail.
- Supervised learning trains a model to predict the "correct" output for a given input. It applies to tasks that require some degree of accuracy relative to some external "ground truth," such as classification or regression.
- Unsupervised learning trains a model to discern intrinsic patterns, dependencies and correlations in data. Unlike in supervised learning, unsupervised learning tasks don't involve any external ground truth against which its outputs should be compared.
- Reinforcement learning (RL) trains a model to evaluate its environment and take an action that will garner the greatest reward. RL scenarios don't entail the existence of a singular ground truth, but they do entail the existence of "good" and "bad" (or neutral) actions.







Applications of Machine Learning

Machine Learning is used in many industries to solve problems and improve services. Here are some common real-world applications:

1. **Healthcare:** It helps doctors to diagnose diseases from medical images like X-rays and MRIs. It also predicts patient outcomes and personalizes treatments which improves healthcare quality.
2. **Finance:** In finance it detects fraudulent transactions in real time and supports algorithmic trading. It also helps to assess credit risk helps in making lending safer and faster.
3. **Retail and E-Commerce:** It helps in personalized product recommendations and forecasts demand to optimize inventory and also analyzes customer sentiment to improve shopping experiences.
4. **Transportation and Automotive:** Self-driving cars rely on ML to navigate and make decisions. It optimizes delivery routes and predicts vehicle maintenance needs which reduces downtime.
5. **Social Media and Entertainment:** Platforms like Netflix and YouTube use ML to recommend content we'll enjoy. It enables image and speech recognition for better user interaction.
6. **Manufacturing:** It improves quality control by detecting defects in products automatically and predicts machine failures in advance and helps in production processes.

Machine learning continues to evolve which helps in opening new possibilities and transforming industries by helping smarter, data-driven decisions and automation which was not possible earlier.

5. Introduction to IoT

The Internet of Things (IoT) connects everyday physical objects—equipped with sensors, software, and network connectivity—to the internet, allowing them to collect, exchange, and act on data, creating smart, responsive environments from homes to industries, enhancing efficiency, automation, and convenience. It's a vast network of "things" (devices, appliances, vehicles) that communicate, extending internet capabilities beyond traditional devices, enabling remote monitoring, data-driven decisions, and new forms of interaction.

Core Concepts

- **"Things":** Physical objects, from smartwatches and thermostats to industrial machines, embedded with technology.
- **Connectivity:** Devices communicate via the internet or other networks.
- **Sensors & Actuators:** Sensors gather data (temperature, motion), while actuators perform actions (adjusting lights).
- **Data Exchange:** Devices share information with each other and systems.
- **Intelligence:** AI/ML helps analyze data for smart, autonomous decisions (e.g., smart home automation).

How It Works

1. **Data Collection:** Sensors on devices gather environmental or operational data.
2. **Data Transmission:** Data is sent over networks (Wi-Fi, cellular).
3. **Data Processing:** Cloud platforms or edge devices analyze the data.
4. **Action/Insight:** Systems respond (e.g., adjust temperature) or provide insights (e.g., crop watering needs).

Examples & Applications

- **Smart Homes:** Connected lights, thermostats, security systems.
- **Healthcare:** Remote patient monitoring, smart wearables.
- **Agriculture:** Soil moisture sensors for efficient irrigation.
- **Industry (IIoT):** Predictive maintenance on factory machinery, smart logistics.
- **Smart Cities:** Traffic management, waste monitoring.

Key Characteristics

- **Interoperability:** Devices use standard protocols to communicate.
- **Self-Configuration:** Systems adapt to changing contexts.
- **Unique Identity:** Each device has a unique identifier.

Why It Matters?

IoT bridges the physical and digital worlds, enabling unprecedented automation, efficiency, and new services, transforming how we live and work by making everyday objects "smart" and interconnected.

Relationship among Bigdata, Machine Learning and 'Internet of Things'

IoT generates massive, real-time data streams, making it a primary source for Big Data; Big Data technologies provide the tools to store and process this influx, while Machine Learning (ML) acts as the "brain," analyzing the complex data to find patterns, predict outcomes, and enable autonomous, intelligent decisions, creating a powerful synergy for automation, efficiency, and insights.

The Interdependent Relationship

1. **IoT: The Data Source**
    - IoT devices (sensors, cameras, smart meters) continuously collect vast amounts of diverse data (volume, velocity, variety) from the physical world.
    - This constant stream of information directly feeds the Big Data ecosystem.
2. **Big Data: The Foundation**
    - Big Data technologies (storage, processing frameworks) are essential to handle the sheer scale and speed of IoT data, which overwhelms traditional systems.
    - It transforms raw IoT data into manageable, usable datasets.
3. **Machine Learning: The Intelligence**
    - ML algorithms analyze these large Big Data sets from IoT to uncover hidden patterns, trends, and anomalies.
    - It enables predictive maintenance (forecasting equipment failure), real-time anomaly detection (smart grids), personalized experiences, and autonomous actions (self-driving cars).

How They Work Together (A Cycle)

- **Collect:** IoT sensors gather data (e.g., temperature, location, usage).
- **Store & Process:** Big Data platforms collect and manage this data.
- **Analyze & Predict:** ML algorithms learn from the data to build models and make predictions or classifications.
- **Act:** The insights from ML drive automated actions or informed human decisions, often back to the IoT devices, creating a closed loop.

Examples in Action

- **Smart Cities:** Traffic sensors (IoT) generate data (Big Data) analyzed by ML to optimize traffic lights in real-time.
- **Manufacturing (Industry 4.0):** Machine sensors (IoT) send data (Big Data) to ML models that predict maintenance needs, preventing costly downtime.

In essence, IoT provides the eyes and ears, Big Data provides the memory and processing power, and Machine Learning provides the intelligence to understand and act upon the world's connected data.